

Strategy Tripod

The first geometric sorting of blocks discards that signal and replaces it with a guess about columns. On two-column pages the guess fails at every figure, every table, and every footnote. We therefore treat stream order as authoritative and only intervene where a page has no text layer at all. Running heads are the second source of noise, because they repeat on every page and sit exactly where a heading would.

Repetition can, since a header recurs across pages while a title appears once. The benchmark documents in this study were built to exercise those two decisions and nothing else. Each document is short, synthetic, and free of copyright, so it can be redistributed with the converter. Hyphenation across a column break is a small case that reveals whether continuation logic inspects the trailing character of a line.

Two Core Propositions

As part of tables were drawn with visible ruling lines so that the vector-based detector fires before the text-alignment fallback. Footnotes were set two points smaller than the body and anchored in the bottom fifth of the column. Their labels are superscript digits, and matching digits appear in the body at the point of reference. The abstract runs across the full measure while the body is set in two columns, which is the arrangement most journals use.

The text-layer path handles digital publications; a raster copy of the same file drives the recognition path. Both paths converge on the same block structure before any processor runs. Nothing in the pipeline depends on a downloaded model, which is the constraint that motivated the project. Line heights are clustered to recover heading levels when a document has no section numbers.